

## **Miniaturization of a Device to Measure Blood Clot Retraction Forces**

Every year an estimated 100,000–300,000 Americans die from thrombotic events, and roughly one in ten in-hospital deaths is linked to pulmonary embolism—many of which are preventable with faster, more precise assessment of coagulation status. Current point-of-care viscoelastic assays such as rotational thromboelastometry (ROTEM) and thromboelastography (TEG) provide valuable whole-blood readouts of clot initiation, strengthening, and lysis, yet they require ~30 min, milliliter-scale samples, and bulky instrumentation. Because platelet–surface interactions govern early hemostasis, we posit that a micro-scale platform that directly reads clot-retraction force can deliver actionable results in minutes from a finger-stick volume, enabling rapid triage in trauma bays, operating rooms, and low-resource settings.

Our long-term goal is to translate microsystems engineering advances into tools that personalize anticoagulation and transfusion therapy. The objective of this proposal is to engineer and validate a chip-scale force-sensing device that quantifies clot formation and breakdown dynamics. Our central hypothesis is that platelet-driven retraction generates pico- to nano-Newton forces that can be captured with microfabricated flexible beams; these force traces will correlate with clinically relevant hemostatic parameters (e.g., clotting time, maximum amplitude, fibrinolysis rate). The rationale is that miniaturizing clot retraction testing will (i) shorten turnaround to <10 min, (ii) reduce sample volume greater than 50-fold, and (iii) open new basic-science windows on platelet biomechanics.

We will pursue this goal via two Specific Aims:

1. Investigate the impact of fibrinogen density on clot-retraction force dynamics using a micro-beam sensor platform. We will fabricate force-sensing microdevices coated with varying fibrinogen concentrations (low, physiological, high). Using donor whole blood, we will quantify key force-trace metrics—time to 10% maximal force (Ct10), peak retraction force (Fmax), and percent force decay at 30 min (Ly30-F)—to assess how fibrinogen density modulates clot retraction. This will define device sensitivity and dynamic range requirements.
2. Optimize micro-beam geometry and material properties to enhance sensitivity to clotting-induced forces. Using standard MEMS fabrication, we will prototype sensors with varying beam widths (100–200  $\mu\text{m}$ ) and materials (e.g., SU-8 vs. PDMS-based composites). Beam deflection during clot formation will be compared across geometries to identify the configuration offering optimal sensitivity without compromising robustness. Results will inform tradeoffs in compliance and manufacturability for disposable use.

**Expected Outcomes and Impact.** We anticipate (i) a robust, chip-based force sensor operable with <40  $\mu\text{L}$  blood and (ii) a validated force-metric panel aligned with existing viscoelastic readouts. Innovation stems from translating platelet biomechanics into an integrated microdevice, shifting the readout from bulk torque to microscale traction forces. This work will lay the foundation for disposable cartridges and portable readers, enabling rapid decision-making in trauma care and anticoagulant management, while providing a versatile research tool for probing clot biophysics in genetic or pharmacologic models.

## Significance

Thrombotic disorders are the third most common cardiovascular condition after heart attack and stroke, and they represent a major public health challenge. In the United States alone, an estimated 100,000 to 300,000 people die annually from blood clots, with most cases occurring in or after hospitalization, making venous thromboembolism (VTE) the leading preventable cause of in-hospital death [1]. There is an urgent need for fast, reliable coagulation assessments to inform treatment in trauma, surgery, and critical care settings.

Current clinical diagnostics such as thromboelastography (TEG) and rotational thromboelastometry (ROTEM) are widely used to evaluate clot formation and breakdown dynamics. While TEG and ROTEM are widely used, they require large blood volumes, have long turnaround times (~30 minutes), and rely on bulky, maintenance-intensive equipment. A review published in the *World Journal of Emergency Surgery* found that the two devices use different reagents and can generate conflicting transfusion recommendations, highlighting the need for better standardization [3].

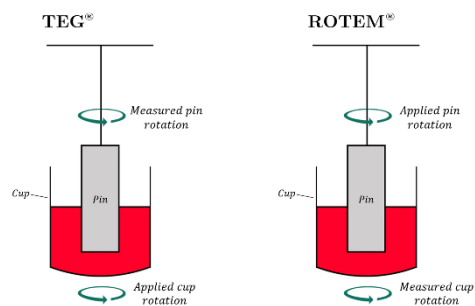


Figure 1: TEG/ROTEM Operating Schematic

While TEG and ROTEM provide useful global assessments of coagulation, neither measures the mechanical forces generated during clot retraction—a process driven by platelet contractility and essential to clot stability. This represents a key knowledge and diagnostic gap. Quantifying clot-retraction forces from microscale blood volumes may enable faster point-of-care diagnostics and open new insights into clot mechanics.

## Innovation

This proposal introduces a novel microfabricated platform for measuring clot retraction forces using a finger-stick volume of whole blood. Unlike conventional viscoelastic tests such as TEG and ROTEM, which rely on bulk mechanical rotation to infer clot properties, our device directly quantifies the traction forces generated by platelets during clot contraction—a biomechanical signal more closely tied to clot integrity and stability. The innovation lies in translating this measurement into a chip-scale, low-cost, and disposable format, enabling point-of-care use without the need for complex instrumentation or large sample volumes.

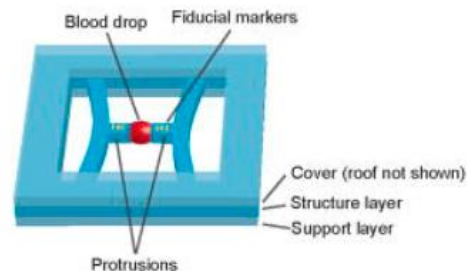


Figure 2: Proposed Device

The system leverages MEMS-based force sensors with tunable geometry and surface coatings to achieve high sensitivity and repeatability. This approach enables precise interrogation of clot mechanics based on platelet–substrate interactions, offering a potentially more informative readout than torque-based methods. By combining low-volume requirements, miniaturization, and scalable fabrication, the device has the potential to democratize viscoelastic testing and advance both clinical diagnostics. Future applications could include a phone-camera based sensing platform for diagnostics in unconventional care settings.

Aim 1: Investigate the impact of fibrinogen density on clot-retraction force dynamics

Beam deflection will be quantified using automated image analysis, and displacement data will be converted to force using beam bending mechanics. Key metrics will include:

- We hypothesize that increasing fibrinogen density will decrease Ct10, increase Fmax, and reduce Ly30-F due to enhanced fibrin matrix stiffness and platelet-fibrin coupling.

To determine the optimal device configuration for detecting clot-induced forces, we will fabricate two sets of sensors: one using SU-8 and one using PDMS. Each set will include beams with widths of 100, 125, 150, 175, and 200  $\mu\text{m}$ . These geometries reflect a tradeoff between bending stiffness and robustness. Devices will be fabricated on two chips—one per material—using standard microfabrication techniques. The timeline in Figure 4 shows the expected fabrication time.

We expect narrower beams to exhibit greater deflection due to lower flexural rigidity, enhancing sensitivity to weak clotting forces. However, excessively compliant beams may suffer from instability or buckling. If buckling is observed, follow-up testing will explore intermediate geometries or alternative materials (e.g., hybrid PDMS-SU-8 structures). Future work may also include the incorporation of integrated strain sensors or alternative coagulant/anticoagulant coatings.

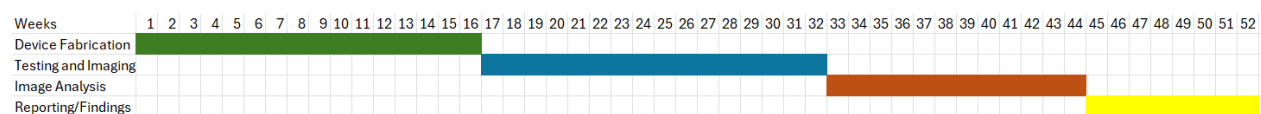


Figure 4: Project Timeline

## References

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2. National Institutes of Health. Venous Thromboembolism. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023 Jan–. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK549877/>
3. Sankarankutty A, Nascimento B, da Luz LT, Rizoli S. TEG® and ROTEM® in trauma: similar test but different results? World Journal of Emergency Surgery. 2012;7(Suppl 1):S3. doi:10.1186/1749-7922-7-S1-S3
4. Lam WA, Chaudhuri O, Crow A, Webster KD, Li T-D, Kita A, Huang J, Fletcher DA. Mechanics and contraction dynamics of single platelets and implications for clot stiffening. Nature Materials. 2011;10(1):61–66. doi:10.1038/nmat2903. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4980575/>

**Rebuttal:****Common Reviewer Concerns:**

The reviewers expressed that the proposal would benefit from additional figures for more clarity in both the current TEG/ROTEM devices, and the proposed new device. There were additional comments about the timeline, formatting of citations, and future directions and implementation of the device and research.

**Reviewer 0:**

This reviewer wanted more discussion on future applications, so a statement was added under the Innovation section to address this feedback.

**Reviewer 1:**

This reviewer wanted more detail on the current methods, so a schematic was included to clarify how they function. The reviewer also had some concerns over the timeline, so the timeline and Gantt chart were updated, along with the citation formatting.

**Reviewer 2:**

The reviewer expressed that the proposed experiment was unclear, so a schematic of the beam bending was added for clarity.

**Instructor:**

The instructor wanted to see a schematic of the device, so to address this feedback it was added. A Gantt chart was added for the timeline instead of a narrative description.